Yuxuan Liu

(+86) 189 6516 9125 · liuyx2005@sjtu.edu.cn · GitHub @Osc-7

Education

Shanghai Jiao Tong University (SJTU), *Undergraduate* — Sept. 2024 - June 2028

B.S. in Computer Science and Technology (Zhiyuan Honors Program, John Hopcroft Class)

- **Honors**: Zhiyuan Honors Program (Top 10% of the university), Zhiyuan Honors Scholarship (2024, 2025), Academic Excellence Scholarship (2025). Expected graduation in June 2028.
- **Selected Coursework**: Programming and Data Structures-III (4.3 / 4.3), Programming Language Design and Implementation (4.3 / 4.3), Computer Systems (3.7 / 4.3).

Publications & Manuscripts

- **Multi-Block Diffusion Language Models**
  Yijie Jin, Jiajun Xu, **Yuxuan Liu**, Chenkai Xu, Jiajun Li, Kai Yu, Pengfei Liu, Zhijie Deng.
  Submitted to EMNLP 2026 (ACL Rolling Review). *Under review.*

Research Experience

diffulex: dLLM Serving Engine, *Undergraduate Research Assistant* — Apr. 2026 - Present

- Participating in research on efficient serving systems for diffusion language models, currently focusing on runtime execution, inference engine evaluation, and serving performance analysis.
- Benchmarked the inference engine across different workloads and runtime settings, and fixed bugs in the serving pipeline to improve model's performance.

Projects

macchiatoBot (LLM Agent Framework) — Nov. 2025 - Present

- Designed and implemented an OS kernel-like architecture to decouple Agent reasoning from tool execution, permission management, and lifecycle control, enabling concurrent multi-agent workflows and multi-agents abilities.
- Developed a centralized Scheduler and TTL-based session eviction mechanism to support long-running processes and cross-terminal session persistence.
- Built an extensible toolchain and memory system featuring working-memory compression and long-term context retrieval strategies.
- Supported multi-user execution with isolated workspaces and concurrent session management; deployed on SJTU's Shiyuan Forum with over 100 active users chatting with it.

Schemer (Scheme-to-x86-64 Compiler) — July 2025

- Developed an x86-64 compiler for a subset of the Scheme language from scratch using Scheme, achieving full self-hosting (bootstrapping).
- Implemented lexical analysis and parsing, and applied optimization techniques (e.g., jump simplification, call optimization) to enhance the performance and readability of the generated assembly code.

Awards & Honors

- **4th Place**, The 1st SJTU HPC Comprehensive Ability Competition "HelloHPC", 2025

Leadership & Others

- **Head of Music Department**, SJTU Art Center — July 2025 - Present